



Modelling Financial Markets

Seminar 16: Digital Assets and DeFi — Hands-on



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Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model)
- ▣ BNB: BNB (native token of BNB Chain, formerly Binance Coin)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ CMKT: Crypto Market factor (value-weighted market excess return)
- ▣ CMOM: Crypto Momentum factor (past winners minus losers)
- ▣ CSMB: Crypto Size factor (small minus big coins)
- ▣ DTB3: 3-Month Treasury Bill: secondary market rate (FRED series code)
- ▣ ETF: Exchange-Traded Fund
- ▣ ETHA: iShares Ethereum Trust ETF (ticker)
- ▣ FRED: Federal Reserve Economic Data
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ IA: Internet Appendix (of the cited paper)
- ▣ KPSS: Kwiatkowski–Phillips–Schmidt–Shin test
- ▣ LVR: Loss-Versus-Rebalancing
- ▣ OLS: Ordinary Least Squares
- ▣ SE: Standard Error
- ▣ US: United States
- ▣ USDC: USD Coin (stablecoin issued by Circle)
- ▣ XRP: XRP (the native token of the XRP Ledger)

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Questions, AI Critique, Project Seed

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the no-arbitrage band, the optimal arbitrage and the LVR of constant-product and weighted pools
- ▣ Model a stablecoin peg as a threshold AR and test linearity with a bootstrap sup-test
- ▣ Test for a break at an unknown date and choose between level and difference estimators
- ▣ Report every estimate with a valid interval: block bootstrap, HAC, Dimson, Forbes–Rigobon

- ▣ Background: Lecture 16; textbooks [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight derivations (A1–A8)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** an open question, a seed for the team project (C1), and an AI answer to critique (C2) and a project seed from the lecture case study (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Crypto trades 7 days a week: annualise with 365 days; joint analyses with shares use common days (prices joined first)
- Report every estimate with its uncertainty: bootstrap interval or robust standard error

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (index and divisor), A3 (fee band and optimal arbitrage), A5 (weighted-pool impermanent loss), A7 (threshold AR algebra)
- Part B: B1 (Bitcoin volatility and the weekend effect), B3 (USD Coin threshold AR), B5 (IBIT: Dimson beta and fee drift), B7 (Strategy: break at an unknown date)

Proposed: you solve them

- Part A: A2 (replacing a constituent; model: A1), A4 (arbitrage in a weighted pool; model: A3), A6 (LVR by $\text{It}\hat{o}$; model: A5), A8 (what a linear AR(1) estimates; model: A7)
- Part B: B2 (Ether and Solana; model: B1), B4 (Tether and Dai; model: B3), B6 (ETHA; model: B5), B8 (Coinbase; model: B7)
- Part C: C1, open question and project seed; C2, critique of an AI answer (Sharpe ratio and stablecoins); C3, crypto market, size and momentum after July 2020 (model: B5, B7)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter16_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- Series: Bitcoin, Ether, Solana; Tether, USD Coin, Dai; IBIT, ETHA, Coinbase, Strategy; S&P 500, Nasdaq 100 (QQQ), gold (GLD), long Treasuries (TLT); daily closes up to 18 September 2026
- Crypto: 7 days a week; shares and ETFs: weekdays, prices adjusted for dividends and splits
- Weekly returns: Friday closes on common days, then log returns

Formulas Used Today

- ▣ Laspeyres index: $I_t = \sum_i P_{i,t} Q_{i,\tau} / D_\tau$; at a reallocation $D_{\tau+1} = \sum_i P_{i,t} Q_{i,\tau+1} / I_t$
- ▣ Constant product $xy = k$, fee f , $\gamma = 1 - f$: $\Delta y = y\gamma\Delta x / (x + \gamma\Delta x)$ Adams, Zinsmeister & Robinson (2020)
- ▣ Impermanent loss $IL(r) = 2\sqrt{r}/(1+r) - 1$; loss versus rebalancing $\sigma^2/8$ a year Milionis et al. (2022)
- ▣ Arbitrage band of a redeemable token: $1 - c_r \leq P \leq 1 + c_m$ Lyons & Viswanath-Natraj (2023); threshold AR with band c Balke & Fomby (1997), linearity test Hansen (1996)
- ▣ Break at an unknown date: sup-Wald Andrews (1993), CI for the date Bai (1997); moving-block bootstrap with blocks of 20–21 days Künsch (1989)
- ▣ Hedge and safe haven Baur & Lucey (2010): $r_t^{\text{BTC}} = a + b r_t^{\text{S\&P}} + \sum_q c_q r_t^{\text{S\&P}} 1\{r_t^{\text{S\&P}} \leq q\} + u_t$

A1: A Three-Coin Index and Its Divisor [Solved]

Task

▣ Data and assumptions

- ▶ Three coins A, B, C on the base day: prices $P_0 = (60\,000; 3\,000; 150)$ USD and circulating supplies $Q = (19.5; 120; 500)$ million
- ▶ Laspeyres index $I_t = \sum_i P_{i,t} Q_{i,\tau} / D_\tau$ with quantities fixed within the month (lecture, Section 3)

▣ Tasks

- ▶ (a) Compute the market values, the weights and the divisor D_0 that sets $I_0 = 1000$.
- ▶ (b) At month end $P_1 = (66\,000; 2\,700; 180)$: compute I_1 and check it as a weighted average of the three returns.
- ▶ (c) At the reallocation the supplies become $(19.52; 120.1; 560)$ million: compute the new divisor D_1 and the new weights.

▣ Report

- ▶ D_0 , I_1 , D_1 and both sets of weights, with every intermediate product

▣ Interpretation question

- ▶ Why must the divisor change at a reallocation, and what would an investor see without the change?

A1: A Three-Coin Index and Its Divisor (Solution 1/2) [Solved]

- (a) Market values $M_i = P_{i,0} Q_i$: 1 170, 360, 75 billion USD, total 1 605 billion.
- Weights $w_i = M_i / \sum_j M_j$: 72.9%, 22.4%, 4.7%; divisor $D_0 = \sum_j M_j / 1000 = 1.605 \cdot 10^9$.
- (b) $I_1 = \sum_i P_{i,1} Q_i / D_0 = 1059.81$; the returns are +10%, -10%, +20%.
- Check: $72.9\% \cdot 10\% - 22.4\% \cdot 10\% + 4.7\% \cdot 20\% = 5.98\% = I_1 / 1000 - 1$.

A1: A Three-Coin Index and Its Divisor (Solution 2/2) [Solved]

- (c) New basket value at month-end prices: $\sum_i P_{i,1} Q_i^{\text{new}}$; the new divisor keeps the level:
 $D_1 = \sum_i P_{i,1} Q_i^{\text{new}} / I_1 = 1.6167 \cdot 10^9$.
- New weights: 75.2%, 18.9%, 5.9%; the level stays 1059.81 on the switch day.
- Interpretation: the divisor absorbs changes in the basket that are not price moves; without it the index would jump by the value of the added coins, a fake return.

A2: Replacing a Constituent [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ The index of A1 after the reallocation in (c): level $I = 1059.81$, divisor $1.6167 \cdot 10^9$
- ▶ On the same day coin C is delisted and coin D enters with $P_D = 0.5$ USD and $Q_D = 40$ billion coins

□ Tasks

- ▶ (a) Compute the market value of the old basket and of the new basket on the switch day.
- ▶ (b) Derive the new divisor that keeps I continuous and compute it.
- ▶ (c) Compute the index level the day after if the divisor were not changed and prices did not move.

□ Report

- ▶ Both basket values, the new divisor and the level of (c)
- ▶ Model: A1

□ Interpretation question

- ▶ Why does the divisor fall here, and who would be misled by a fake loss?

A3: The No-Arbitrage Band of a Pool with Fees [Solved]

Task

□ Data and assumptions

- ▶ Constant-product pool $xy = k$ with $x = 1\,000$ Ether and $y = 3\,000\,000$ USD Coin, so the pool price is $p = y/x = 3\,000$ Adams, Zinsmeister & Robinson (2020)
- ▶ Fee $f = 0.3\%$ charged on the input, $\gamma = 1 - f$; the outside market price is P ; no gas cost, one trade

□ Tasks

- ▶ (a) Derive the marginal cost of buying and the marginal revenue of selling Ether in the pool, and show that no trade is profitable while $\gamma P \leq p \leq P/\gamma$.
- ▶ (b) For $P = 3\,300$ write the profit $\pi(\Delta y)$ of buying Ether with Δy USD Coin, solve the first-order condition and compute Δy^* ; the fee stays in the pool (Uniswap v2): compute the actual reserves and the new pool price y/x .
- ▶ (c) Compute the arbitrage profit with and without the fee.

□ Report

- ▶ The band, Δy^* , Δx , the new price and both profits

□ Interpretation question

- ▶ Where does arbitrage leave the pool price, and who pays the arbitrage profit?

A3: The No-Arbitrage Band of a Pool with Fees (Solution 1/2) [Solved]

- (a) Buying Δx Ether: $(x - \Delta x)(y + \gamma \Delta y) = k$ gives $\Delta y = \frac{1}{\gamma} \left(\frac{k}{x - \Delta x} - y \right)$, so the marginal cost at $\Delta x = 0$ is $\frac{k}{\gamma x^2} = p/\gamma$.
- Selling Ether: $(x + \gamma \Delta x)(y - \Delta y) = k$ gives a marginal revenue γp .
- Buying pays iff $p/\gamma < P$, selling pays iff $\gamma p > P$; neither holds when $\gamma P \leq p \leq P/\gamma$, here $[3290.10, 3309.93]$ for $P = 3\,300$.

A3: The No-Arbitrage Band of a Pool with Fees (Solution 2/2) [Solved]

- (b) $\pi(\Delta y) = P[x - k/(y + \gamma\Delta y)] - \Delta y$; $\pi'(\Delta y) = Pk\gamma/(y + \gamma\Delta y)^2 - 1 = 0$ gives the effective reserve $y^e = y + \gamma\Delta y^* = \sqrt{\gamma Pk}$.
- Substituting $k = 3 \cdot 10^9$, $\gamma = 0.997$, $P = 3300$: $y^e = 3141703$, $x' = k/y^e = 954.90$; $\Delta y^* = (y^e - y)/\gamma = 142130$ USD
Coin buys $\Delta x = 45.10$ Ether.
- The fee stays in the pool: actual reserves x' and $y' = y + \Delta y^* = 3142129.75$, so the new pool price is $p' = y'/x' = 3290.55$, above $y^e/x' = \gamma P = 3290.10$; k grows with the retained fee.
- (c) Profit $P\Delta x - \Delta y^* = 6713$ USD; without the fee 7147 USD.
- Interpretation: arbitrage stops when the marginal cost of Ether in the pool reaches P ; the pool price ends inside its new band, $P \in [\gamma p', p'/\gamma] = [3280.67, 3300.45]$, not at P ; the profit is paid by the liquidity providers, who sold Ether below the market price.

A4: Arbitrage in a Weighted Pool [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Weighted pool (Balancer type) $x^w y^{1-w} = k$ with $w = 0.8$, $x = 1\,000$ Ether and pool price $p = 3\,000$; fee 0.3% on the input
- ▶ The outside price rises to $P = 3\,300$; one arbitrage trade, no gas cost

□ Tasks

- ▶ (a) Differentiate the invariant to show that the pool price is $p = \frac{w}{1-w} \frac{y}{x}$, and compute y .
- ▶ (b) Show that the no-arbitrage band $\gamma P \leq p \leq P/\gamma$ does not depend on w .
- ▶ (c) Derive the reserves after the optimal arbitrage (the fee stays in the pool, as in A3), compute Δy^* , the profit and the profit per unit of pool value (the pre-trade value $px + y$ at the pool price; A3: 6 million USD), and compare with A3.

□ Report

- ▶ y , x' , the effective and actual y' , Δy^* , the pool value, the profit, the profit per unit of value and the ratio to A3
- ▶ Model: A3

□ Interpretation question

- ▶ Why does a pool with $w = 0.8$ lose less to arbitrageurs than a 50/50 pool?

A5: Impermanent Loss of a Weighted Pool [Solved]

Task

▣ Data and assumptions

- ▶ A liquidity provider owns a pool $x^w y^{1-w} = k$; arbitrage keeps the pool at the market price P (A4, without fees)
- ▶ The price moves from P_0 to $P_1 = rP_0$; the benchmark is holding the initial tokens

▣ Tasks

- ▶ (a) Show that the value shares stay w and $1 - w$ and that $V(P) = cP^w$.
- ▶ (b) Derive $IL_w(r) = r^w / (wr + 1 - w) - 1$ and check that $w = 1/2$ gives $2\sqrt{r}/(1 + r) - 1$.
- ▶ (c) Expand $\ln(1 + IL_w)$ to second order in $u = \ln r$ and compute IL_w for $r = 2$ and $r = 4$, $w = 0.5$ and 0.8 .

▣ Report

- ▶ The formula, the approximation and a table of four values

▣ Interpretation question

- ▶ Which pool loses less when the price doubles, and why?

A5: Impermanent Loss of a Weighted Pool (Solution 1/2) [Solved]

- (a) From A4(a) the pool price equals P : $Px = \frac{w}{1-w}y$, so $Px = wV$ and $y = (1-w)V$ with $V = Px + y$.
- Substituting into the invariant: $(wV/P)^w((1-w)V)^{1-w} = k$, so $V = k w^{-w}(1-w)^{w-1}P^w = cP^w$.
- (b) Pool: $V_1/V_0 = r^w$; holding: $V_H/V_0 = wr + (1-w)$; so $IL_w(r) = r^w/(wr + 1 - w) - 1$; at $w = 1/2$: $\sqrt{r}/[(r+1)/2] - 1 = 2\sqrt{r}/(1+r) - 1$.

A5: Impermanent Loss of a Weighted Pool (Solution 2/2) [Solved]

- (c) $\ln(1 + \text{IL}_w) = wu - \ln(1 + w(e^u - 1))$; with $e^u - 1 = u + u^2/2$ and $\ln(1 + z) = z - z^2/2$: $\ln(1 + \text{IL}_w) \approx -\frac{1}{2}w(1 - w)u^2$.
- $r = 2$: -5.72% for $w = 0.5$ (approximation -6.01%) and -3.27% for $w = 0.8$ (-3.84%).
- $r = 4$: -20.00% ($w = 0.5$) and -10.84% ($w = 0.8$); the approximation is good for moderate moves only.
- Interpretation: the loss scales with $w(1 - w)$, the curvature of $V(P)$; a pool that holds mostly Ether rebalances less against the trend and sells less volatility.

A6: Loss Versus Rebalancing by Itô [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Price $dP_t = \mu P_t dt + \sigma P_t dW_t$; pool value $V(P) = cP^w$ (A5); frictionless arbitrage, no fees [Milionis et al. \(2022\)](#)
- ▶ The rebalancing portfolio holds at every instant the same quantity of Ether as the pool, $V'(P_t)$, and trades at market prices

□ Tasks

- ▶ (a) Compute V' and V'' , apply Itô's lemma to $V(P_t)$, and show that $d(R_t - V_t) = \frac{1}{2}w(1-w)\sigma^2 V_t dt$.
- ▶ (b) Check that $w = 1/2$ gives the lecture's rate $\sigma^2/8$.
- ▶ (c) Compute the annual LVR rate for $\sigma = 70\%$ and $w = 0.5, 0.8$ and 0.95 .

□ Report

- ▶ The derivation and three rates in % a year
- ▶ Model: A5

□ Interpretation question

- ▶ Why does the drift μ not appear in the loss?

A7: Threshold AR Algebra [Solved]

Task

▣ Data and assumptions

- ▶ Peg deviation in basis points: $d_t = \phi_{\text{in}} d_{t-1} + \varepsilon_t$ if $|d_{t-1}| \leq c$, and $d_t = \phi_{\text{out}} d_{t-1} + \varepsilon_t$ otherwise Balke & Fomby (1997)
- ▶ USD Coin estimates from B3: $c = 2.8650$, $\phi_{\text{in}} = 0.5855$, $\phi_{\text{out}} = 0.2832$; use the skeleton $\varepsilon_t = 0$

▣ Tasks

- ▶ (a) Derive the half-life in each regime and compute it.
- ▶ (b) Starting from $d_0 = -285$ bp (the close of 11 March 2023), derive and compute the number of days until the deviation is back inside the band.
- ▶ (c) State what $\phi_{\text{in}} = 1$ would mean economically.

▣ Report

- ▶ Two half-lives in days and the number of days of (b)

▣ Interpretation question

- ▶ What does $\phi_{\text{in}} < 1$ say about who trades inside the band?

A7: Threshold AR Algebra (Solution) [Solved]

- (a) While the process stays in one regime, $d_h = \phi^h d_0$; $\phi^h = 0.5$ gives $h_{1/2} = \ln 0.5 / \ln \phi$.
- Inside: $\ln 0.5 / \ln 0.5855 = 1.29$ days; outside: $\ln 0.5 / \ln 0.2832 = 0.55$ days.
- (b) Outside the band $|d_n| = |d_0| \phi_{\text{out}}^n \leq c \iff n \geq \ln(c/|d_0|) / \ln \phi_{\text{out}} = \ln(2.8650/285) / \ln 0.2832 = 3.65$.
- So after 4 days the deviation is inside the band; from then on the slower inside dynamics apply.
- (c) $\phi_{\text{in}} = 1$ is a random walk inside the band, the band-TAR case of [Balke & Fomby \(1997\)](#): no one trades until arbitrage pays.
- Interpretation: the estimate $\phi_{\text{in}} < 1$ means market makers pull the price towards par even inside the band.

A8: What Does a Linear AR(1) Estimate? [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ The USD Coin deviations of A7 and B3; a linear AR(1) $d_t = \phi d_{t-1} + e_t$ fitted by OLS without an intercept
- ▶ $\hat{\phi}_{\text{in}}$ and $\hat{\phi}_{\text{out}}$ are the regime-wise OLS estimates at the threshold $\hat{c}$ of B3

□ Tasks

- ▶ (a) Show that, in sample, $\hat{\phi}_{\text{lin}} = (1 - s)\hat{\phi}_{\text{in}} + s\hat{\phi}_{\text{out}}$ with $s = \sum_{\text{out}} d_{t-1}^2 / \sum_t d_{t-1}^2$.
- ▶ (b) Compute s for USD Coin and check the identity numerically against $\hat{\phi}_{\text{lin}}$ from B3.
- ▶ (c) Compute the linear half-life and say which regime it describes.

□ Report

- ▶ s , the weighted average, $\hat{\phi}_{\text{lin}}$ and the linear half-life
- ▶ Model: A7

□ Interpretation question

- ▶ Why can a linear AR(1) report “fast reversion” although most days show slow reversion?

B1: Bitcoin's Volatility and the Weekend Effect [Solved]

Task

▣ Data and assumptions

- ▶ Daily log returns of Bitcoin (BTC-USD.CC), 2 January 2018 – 18 September 2026, 3 182 days, 7 days a week; S&P 500 (GSPC.INDX) on its own calendar
- ▶ Split at the first spot ETF day, 11 January 2024: 2 200 days before, 982 after

▣ Tasks

- ▶ (a) Compute the annualised volatility $s\sqrt{365}$ with a 95% moving-block bootstrap interval (blocks of 20 days, 2 000 resamples) [Künsch \(1989\)](#).
- ▶ (b) Compute the ratio of the weekend to the weekday variance of returns before and after the ETFs, each with a block-bootstrap interval (blocks of 21 days).
- ▶ (c) Compute a 95% interval for the change in the ratio from the difference of the two bootstrap distributions.

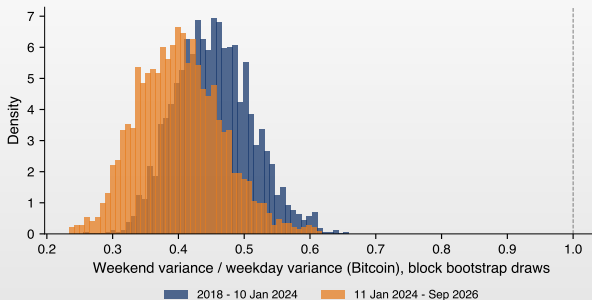
▣ Report

- ▶ Volatility with CI and the ratio to the S&P 500; both ratios with CIs; the change with CI; the bootstrap histogram

▣ Interpretation question

- ▶ Is the weekend lull a new, ETF-driven effect? Why blocks and not single days?

B1: Bitcoin's Volatility and the Weekend Effect (Solution 1/2) [Solved]



- Block-bootstrap distributions of the weekend / weekday variance ratio; the dashed line marks a ratio of 1

B1: Bitcoin's Volatility and the Weekend Effect (Solution 2/2) [Solved]

- (a) Steps: compute s , resample 2 000 series of 20-day blocks, recompute $s\sqrt{365}$, take the 2.5% and 97.5% percentiles.
- Result: 64.1%, 95% CI [58.2, 70.4]; S&P 500 19.2%, a ratio of 3.3; excess kurtosis 14.9.
- (b) Before: 0.45 [0.35, 0.57]; after: 0.39 [0.29, 0.53].
- (c) Change -0.05 , 95% CI $[-0.22, 0.11]$: the interval contains 0, so at the 5% level we do not reject an unchanged weekend effect.
- Interpretation: weekends have less than half the weekday variance before and after 2024, so the lull is not ETF-driven; blocks keep the volatility clusters that single-day resampling would destroy.
- Caveat: these percentile intervals for a variance (and a variance ratio) assume $Er^4 < \infty$; with $\hat{\alpha}_H$ near 2.7 (lecture) this is doubtful and the bootstrap need not be consistent [Athreya \(1987\)](#): treat the intervals as indicative (an m -out-of- n bootstrap is the robust alternative).

B2: Ether and Solana [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ Daily log returns of Ether (ETH-USD.CC, 2018–2026) and Solana (SOL-USD.CC, April 2020 – 2026); S&P 500 over the same periods
- ▶ Same split date (11 January 2024) and bootstrap settings as B1

▣ Tasks

- ▶ (a) Compute each coin's annualised volatility with a block-bootstrap CI and its ratio to the S&P 500 volatility.
- ▶ (b) Compute the weekend / weekday ratio before and after the ETFs and a CI for the change.

▣ Report

- ▶ One table with both coins and Bitcoin from B1
- ▶ Model: B1

▣ Interpretation question

- ▶ Which coin is the riskiest relative to shares? Is the weekend effect weaker for coins with more on-chain activity?

B3: Is the USD Coin Peg a Threshold Process? (1/2) [Solved]

Task

▣ Data and assumptions

- ▶ Daily close of USD Coin (USDC-USD.CC), 1 January 2021 – 18 September 2026, 2 087 days; deviation $d_t = 10^4(P_t - 1)$ in basis points
- ▶ Model of A7 (no intercept); errors may be heteroskedastic; test level 5%

▣ Tasks

- ▶ (a) Compute the standard deviation, the number of days with $|d_t| > 50$ bp, the lowest close and the lowest daily low.
- ▶ (b) Estimate c by least squares over the grid of $|d_{t-1}|$ values that leaves at least 15% of the days in each regime, and estimate $\phi_{\text{in}}, \phi_{\text{out}}$ with White standard errors.
- ▶ (c) Test $H_0: \phi_{\text{in}} = \phi_{\text{out}}$ with $\sup_c W(c)$ and the fixed-regressor bootstrap Hansen (1996): $d_t^* = \hat{e}_t \eta_t$, $\eta_t \sim N(0, 1)$, $\hat{e}_t$ the residuals of the linear AR(1) under H_0 , regressors d_{t-1} fixed, 999 draws; also report the χ_1^2 p -value.
- ▶ (d) Repeat (b)–(c) without 9–16 March 2023: drop every pair (d_{t-1}, d_t) that touches these dates, so that no pair spans the gap (8 and 17 March are not consecutive).

B3: Is the USD Coin Peg a Threshold Process? (2/2) [Solved]

Task

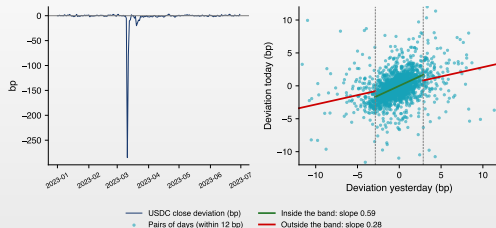
☐ Report

- ▶ $\hat{c}$, both slopes with SE, half-lives, sup W , both p -values, the scatter chart

☐ Interpretation question

- ▶ Why is χ^2_1 wrong here, and does one stress week carry the evidence of a band?

B3: Is the USD Coin Peg a Threshold Process? (Solution 1/3) [Solved]



- Left: close deviation in the first half of 2023; right: today against yesterday within ± 12 bp, with the fitted slopes inside and outside $\pm \hat{c}$ (dashed lines)
- (a) s.d. 7.5 bp (without March 2023: 3.7); 3 days beyond 50 bp; lowest close -285 bp and lowest low -1226 bp, both on 11 March 2023.

B3: Is the USD Coin Peg a Threshold Process? (Solution 2/3) [Solved]

- (b) Steps: sort the days by $|d_{t-1}|$; for each c on the grid $[0.3, 2.9]$ fit OLS in each regime and keep the c with the smallest total sum of squared residuals.
- Result: $\hat{c} = 2.9$ bp, 15.7% of days outside; $\hat{\phi}_{\text{in}} = 0.59$ (SE 0.04), $\hat{\phi}_{\text{out}} = 0.28$ (SE 0.03); half-lives 1.29 and 0.55 days.
- (c) $W(c) = (\hat{\phi}_{\text{in}} - \hat{\phi}_{\text{out}})^2 / (\hat{v}_{\text{in}} + \hat{v}_{\text{out}})$ on each c ; $\sup W = 32.5$.
- Bootstrap: regress d_t^* on the same fixed d_{t-1} , recompute $\sup W^*$ 999 times; $p = 0.001$ (χ_1^2 : < 0.001); decision: reject linearity at 5%.

B3: Is the USD Coin Peg a Threshold Process? (Solution 3/3) [Solved]

- (d) Without the pairs touching 9–16 March 2023: $\sup W = 5.5$, bootstrap $p = 0.291$: linearity is no longer rejected; χ_1^2 would give 0.019, a rejection by a miscalibrated test.
- Why χ_1^2 fails: c is not identified under H_0 , and the supremum over many c is larger than any single $W(c)$ (the Davies problem).
- Why a wild bootstrap: the excess kurtosis is 1031; multiplying each residual by η_t keeps its own variance, which an i.i.d. resampling of residuals would mix across calm and stress days.
- Economic conclusion: the evidence for nonlinear adjustment is sensitive to the stress week; the remaining sample does not reject a linear, fast-reverting AR (non-rejection, not proof of linearity).

B4: Tether and Dai [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ Daily closes of Tether (USDT-USD.CC) and Dai (DAI-USD.CC), 1 January 2021 – 18 September 2026; deviations in bp
- ▶ Hypothesis to discuss: Tether has the widest band, because fewer traders can redeem directly with the issuer; the tests below are tests of linearity, not of differences between bands

▣ Tasks

- ▶ (a) Estimate the threshold AR of B3 for each coin: $\hat{c}$, $\hat{\phi}_{\text{in}}$, $\hat{\phi}_{\text{out}}$ with White SE.
- ▶ (b) Test linearity with sup W and the fixed-regressor bootstrap (999 draws), with and without 9–16 March 2023, at the 5% level.

▣ Report

- ▶ One table with Tether, Dai and USD Coin (B3)
- ▶ Model: B3

▣ Interpretation question

- ▶ Which estimated band is widest, and for which coins does the bootstrap reject linearity? What would a test of the difference between two thresholds need? Which of the three coins reverts fastest outside its band?

B5: IBIT: Tracking, Timing and the Fee Drift [Solved]

Task

▣ Data and assumptions

- ▶ IBIT (IBIT.US, adjusted close) and Bitcoin (BTC-USD.CO) on IBIT's trading days, 11 January 2024 – 18 September 2026: 673 days and 140 Friday-to-Friday weeks; prices joined first, then log returns
- ▶ Sponsor fee 0.25% a year paid in Bitcoin [BlackRock \(2026\)](#); level 5%

▣ Tasks

- ▶ (a) Estimate the daily and the weekly beta with Newey–West SE [Newey & West \(1987\)](#), and the Dimson sum beta $b_{-1} + b_0 + b_1$ with one lag and one lead [Dimson \(1979\)](#).
- ▶ (b) Run an ADF test (with trend, lags by AIC) [Dickey & Fuller \(1979\)](#) and a KPSS test (with trend) [Kwiatkowski et al. \(1992\)](#) on $\ell_t = \ln(P_t^{\text{IBIT}} / P_t^{\text{BTC}})$.
- ▶ (c) Estimate the annual fee drift as the mean of $\Delta \ell_t$ (HAC SE, 6 lags) and as the slope of ℓ_t on time in years (HAC SE, 20 lags), each with a 95% CI.

▣ Report

- ▶ Three betas with SE, the two test statistics with decisions, both drifts with CIs

▣ Interpretation question

- ▶ Why is the daily OLS beta below 1? Which drift estimator is valid and which is precise?

B5: IBIT: Tracking, Timing and the Fee Drift (Solution 1/2) [Solved]

- (a) Daily OLS $\hat{\beta} = 0.93$; weekly $\hat{\beta} = 1.007$ (SE 0.011), $R^2 = 0.98$.
- Dimson: $\hat{b}_{-1} = 0.05$, $\hat{b}_0 = 0.94$, $\hat{b}_1 = 0.01$; sum 1.00 (SE 0.023), $t(\beta = 1) = -0.07$: not different from 1.
- Reading: part of each Bitcoin move reaches IBIT a day later; the lag term restores $\beta = 1$ on daily data, as the weekly regression does.
- (b) ADF = -11.07 , $p < 0.001$: reject the unit root at 5%; KPSS = 0.041, $p > 0.10$: do not reject trend stationarity.

B5: IBIT: Tracking, Timing and the Fee Drift (Solution 2/2) [Solved]

- (c) Mean of $\Delta \ell_t$ times days per year: -0.36% , 95% CI $[-9.28, 8.57]$.
- Trend slope: -0.237% a year, 95% CI $[-0.315, -0.158]$: excludes 0 and contains -0.25% .
- Why they differ: the mean of $\Delta \ell_t$ telescopes to $(\ell_T - \ell_0)/T$, so it uses only two noisy days; the trend slope uses every day.
- Under the maintained model $\ell_t = \text{fee trend} + \text{stationary timing noise}$ (ADF rejects, KPSS does not), the trend slope with HAC SE is consistent and its CI is valid; the mean of $\Delta \ell_t$ is also consistent, but $\Delta \ell_t$ then has zero long-run variance, so its HAC interval is poorly calibrated.
- With a unit root plus drift, the trend slope still estimates the drift, but the usual trend HAC interval is invalid; the mean of differences then has the standard CI. Choosing by a pretest does not guarantee post-selection coverage.
- Economic conclusion: the fee is visible in the data at the 0.25% rate, and IBIT tracks Bitcoin one for one.

B6: ETHA and Ether [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ ETHA (ETHA.US, adjusted close) and Ether (ETH-USD.CC) on ETHA's trading days, 23 July 2024 – 18 September 2026: 541 days, 112 weeks
- ▶ Sponsor fee 0.25% a year; level 5%

▣ Tasks

- ▶ (a) Estimate the daily, weekly and Dimson betas as in B5(a).
- ▶ (b) Run the ADF and KPSS tests of B5(b) on the log price ratio.
- ▶ (c) Estimate both drifts of B5(c) with 95% CIs.

▣ Report

- ▶ Betas with SE, tests with decisions, both drifts with CIs, tracking errors
- ▶ Model: B5

▣ Interpretation question

- ▶ Is ETHA a worse tracker than IBIT, or only noisier? How many years would pin the fee down to $\pm 0.05\%$?

B7: When Did Strategy's Bitcoin Beta Change? (1/2) [Solved]

Task

▣ Data and assumptions

- ▶ Weekly (Friday) log returns of Strategy (MSTR.US, formerly MicroStrategy), Bitcoin and the S&P 500, 16 April 2021 – 18 September 2026, 284 weeks; prices joined first
- ▶ Model $r_t^{\text{MSTR}} = a + \beta_{\text{BTC}} r_t^{\text{BTC}} + \beta_{\text{SPX}} r_t^{\text{SPX}} + u_t$; level 5%

▣ Tasks

- ▶ (a) Estimate the model with Newey–West SE on the full period and on the two subperiods split at the first spot ETF day (weeks up to 5 January 2024; from 12 January 2024), and test $\beta_{\text{BTC}} = 1$ in each.
- ▶ (b) Test for a break in all three coefficients at an unknown date with sup-Wald (HAC covariance in each regime, 15% trimming) [Andrews \(1993\)](#); take the 5% critical value from the simulated limit $\sup_r \|B(r) - rB(1)\|^2 / [r(1-r)]$, B three-dimensional.
- ▶ (c) Estimate the break date $\hat{T}$ by least squares and its 95% CI [Bai \(1997\)](#): $\hat{T} \pm (\lceil 11/\hat{L} \rceil + 1)$ weeks, $\hat{L} = (\hat{\delta}' \hat{Q} \hat{\delta})^2 / (\hat{\delta}' \hat{\Omega} \hat{\delta})$, where $\hat{\delta}$ is the post-minus-pre coefficient vector, $\hat{Q} = X'X/T$, and $\hat{\Omega}$ the Bartlett long-run covariance of $x_t \hat{u}_t$ (Newey–West lags, residuals of the two-regime fit).
- ▶ Assumptions: a small shift, Q and Ω the same in both regimes; 11 is the 97.5% quantile of $\arg \max_s \{W(s) - |s|/2\}$, W a two-sided Brownian motion, so the interval is symmetric.

B7: When Did Strategy's Bitcoin Beta Change? (2/2) [Solved]

Task

☐ Report

- ▶ The table of (a), sup W with the critical value and the decision, the date with its CI and the coefficients in each regime

☐ Interpretation question

- ▶ Do the data select January 2024? Why is a split at an imposed date not a test of when the beta changed?

B7: When Did Strategy's Bitcoin Beta Change? (Solution 1/2) [Solved]

	n	$\hat{\beta}_{BTC}$ (SE)	$t(\beta_{BTC} = 1)$	$\hat{\beta}_{SPX}$ (SE)	alpha, % a year (t)
□ Apr 2021 – Sep 2026	284	1.18 (0.11)	1.64	1.02 (0.24)	−4.5 (−0.19)
□ Apr 2021 – 5 Jan 2024	143	1.01 (0.14)	0.08	1.40 (0.35)	−0.7 (−0.02)
□ 12 Jan 2024 – Sep 2026	141	1.50 (0.12)	4.12	0.48 (0.29)	−9.3 (−0.29)

- (a) The imposed split suggests a change: $\hat{\beta}_{BTC}$ from 1.01 to 1.50 ($t = 4.12$ against 1 after 2024); alpha is never significant.

B7: When Did Strategy's Bitcoin Beta Change? (Solution 2/2) [Solved]

- (b) Steps: for every candidate week in the middle 70% fit both regimes, compute $W(\tau)$ with HAC covariances and take the maximum.
- Result: $\sup W = 35.49$ against a 5% critical value of 13.95, $p < 0.001$: reject stability at 5%.
- (c) Break date 24 Jun 2022, 95% CI 31 Dec 2021 – 16 Dec 2022; W at the ETF launch only 8.64.
- Regimes: Bitcoin beta $0.69 \rightarrow 1.48$; S&P 500 beta $1.25 \rightarrow 1.00$.
- Interpretation: the data choose mid-2022, a period of market-wide crypto stress, not the ETF launch; the date alone does not identify the cause; an imposed split finds a change wherever the true break lies before it, so it cannot date the change.

B8: Coinbase [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Weekly log returns of Coinbase (COIN.US, listed April 2021), Bitcoin and the S&P 500, 16 April 2021 – 18 September 2026, 283 weeks
- ▶ Same model, settings and level as B7

□ Tasks

- ▶ (a) Estimate the two-factor model on the full period and the two subperiods with Newey–West SE.
- ▶ (b) Run the sup-Wald test for a break at an unknown date.
- ▶ (c) Estimate the break date and its 95% CI, and the coefficients in each regime.

□ Report

- ▶ Table of (a), the test with its decision, the date with CI
- ▶ Model: B7

□ Interpretation question

- ▶ Is Coinbase a Bitcoin proxy or a US growth stock? Do Strategy and Coinbase break in the same week, and what common event could explain it?

C1: Alternative Asset, Risk Asset or Hedge? (1/2) [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ Daily log returns of Bitcoin, the S&P 500, Nasdaq 100 (QQQ), gold (GLD) and long Treasuries (TLT) on common days (prices joined first), January 2021 – 10 January 2024 against 11 January 2024 – 18 September 2026
- ▶ Definitions of hedge and safe haven: [Baur & Lucey \(2010\)](#); earlier evidence for Bitcoin: [Bouri et al. \(2017\)](#); level 5%

▣ Tasks

- ▶ (a) Compute the correlations with Bitcoin in both periods, with moving-block bootstrap CIs (20-day blocks, 2 000 draws) for each and for the change.
- ▶ (b) Test $H_0: \Delta\rho = 0$ for the S&P 500 with the Forbes–Rigobon adjusted correlation [Forbes & Rigobon \(2002\)](#) and a block-bootstrap CI.
- ▶ (c) Estimate the hedge / safe-haven regression with interactions at the 10%, 5% and 1% S&P 500 quantiles and Newey–West SE, and the quantile-regression betas $\beta(\tau)$ at $\tau = 1\%, 5\%, 50\%$ [Koenker & Bassett \(1978\)](#) with block-bootstrap SE.
- ▶ (d) Compare downside and upside betas and Bitcoin's return on the ten worst S&P 500 days.

C1: Alternative Asset, Risk Asset or Hedge? (2/2) [Proposed]

Solution discussed in class

Task

☐ Report

- ▶ A correlation table, the test of (b), the regression and quantile betas, a two-slide verdict linked to the distribution features of the lecture ([Pele et al. \(2023\)](#))

☐ Interpretation question

- ▶ Is Bitcoin “alternative” in shape but “risk” in co-movement? Did the ETFs change that?

C2: Critique an AI Answer [Proposed]

- A student asks an AI assistant: “Compare Bitcoin with lending USD Coin (USDC) at 5% a year: compute Bitcoin’s annualised Sharpe ratio on daily data for 2021–2025, with the USDC yield as the risk-free rate”
- The assistant answers with this code:

```
import numpy as np, pandas as pd
btc = pd.read_csv('data/market/BTC-USD.CC.csv', index_col='date',
                  parse_dates=True)['close']
r = np.log(btc.loc['2021':'2025']).diff().dropna()    # daily log returns
mu = r.mean() * 252                                  # annual mean
sd = r.std() * np.sqrt(252)                           # annual volatility
rf = 0.05                                             # USDC lending: risk-free
sharpe = (mu - rf) / sd
print(f"mu={mu:.1%}, vol={sd:.1%}, Sharpe={sharpe:.2f}")
```

- “Output: mu = 15.1%, vol = 48.5%, Sharpe = 0.21. USDC is fully backed by dollars and has never traded below \$0.99, so lending it is a risk-free investment”

C2: Critique an AI Answer, Task [Proposed]

Solution discussed in class

Task

- The answer contains several substantive errors, in the code and in the text: find all of them
- For each error
 - ▶ say what is wrong and why
 - ▶ say how you would detect it: a count in the data, the data themselves, a definition from the lecture
 - ▶ give the corrected line or statement
- Rerun the corrected code (returns computed before selecting 2021–2025, simple excess returns, r_f from Treasury bills); report the mean, the volatility and the Sharpe ratio
- Log the errors as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version

Interpretation

- How many daily returns should 2021–2025 contain for Bitcoin? How many does the code produce, and why?
- Which of the errors can you check in one line with the course data?
- If lending USDC were risk-free, why would it pay more than a Treasury bill?

C3: Crypto Market, Size and Momentum After July 2020 (1/2) [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ Context: [Liu, Tsyvinski & Wu \(2022\)](#) (lecture case study), building on [Liu & Tsyvinski \(2021\)](#), price 1,827 coins with CMKT, CSMB and CMOM (2014 – July 2020); three-week momentum earns 3.1% a week (Table 4), most among large coins [Liu, Tsyvinski & Wu \(2021\)](#)
- ▶ Daily closes of Bitcoin, Ether, XRP, BNB, Cardano, Solana, Dogecoin, Litecoin and Chainlink from data/market; the five-coin market-value index of Lecture 16, Section 3, as the CMKT proxy; the one-month Treasury bill rate and the global five factors [Fama & French \(2015\)](#) from the [Kenneth French Data Library](#); the Canadian, Singapore and Australian dollar, euro and pound against the USD from FRED; in sample up to the end of July 2020, out of sample August 2020 – 18 September 2026; level 5%

▣ Tasks

- ▶ (a) Build weekly returns with the paper's calendar (52 weeks a year, Section 2) for Bitcoin and XRP from the first week of 2014 and for Ether from week 32 of 2015 to July 2020; compare the mean, median, standard deviation, skewness and kurtosis with Table 1, Panel B.
- ▶ (b) Regress the weekly excess return of the CMKT proxy on the global five factors and the five currency returns (footnote 16, Table IA.3), in and out of sample; report the coefficients with HAC t -statistics [Newey & West \(1987\)](#) and the R^2 .

C3: Crypto Market, Size and Momentum After July 2020 (2/2) [Proposed]

Solution discussed in class

Task

Tasks

- ▶ (c) Each week sort the nine coins into terciles by past three-week return (top-20 design, Table IA.13) and hold top minus bottom for one week, equal-weighted (market values are public for five coins only); report its mean, t -statistic and crypto-CAPM α , β , R^2 (Eq. 2, Table 8), in and out of sample.
- ▶ (d) Test whether the momentum alpha of (c) changed after July 2020: an interaction with HAC errors and a sup-Wald test at an unknown date [Andrews \(1993\)](#), as in B7.

Report

- ▶ A table paper – replication for (a); in- and out-of-sample tables for (b) and (c); the break test with its estimated date; the full list of tests run
- ▶ Model: B5, B7

Interpretation question

- ▶ If the momentum alpha of nine large coins stays significant after July 2020, is that evidence for the CMOM factor or for the selection of coins that survived to 2026? Which of your numbers, or which extra data, would separate the two readings?

Common Pitfalls

- ▣ Forgetting the divisor when an index changes constituents or supplies
- ▣ Using the pool price as the execution price: large trades pay the price impact
- ▣ Judging a liquidity position against holding instead of against rebalancing
- ▣ Testing a threshold or a break date with χ^2 critical values; imposing the break date instead of estimating it
- ▣ Daily OLS betas between assets that close at different times (use Dimson); a trend regression without a unit-root test
- ▣ Calling Bitcoin a hedge without testing the extreme-day betas

Take-Home Assignment (Team Project)

- ▣ Choose three crypto-assets not analysed in class (or a stablecoin and two coins)
- ▣ Build a market-value weighted index with a divisor and compare it with Bitcoin: return, volatility, tracking error, with bootstrap intervals
- ▣ Run the hedge / safe-haven regression of Part C for your index against the S&P 500 and gold
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- ▣ Log every AI error you catch in `AI_ERRORS.md`, as in C2: what was asked, what was wrong, how you found out, the correct version

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Are Cryptos becoming alternative Assets?
- ▣  Comparison of Cryptocurrency Indices
- ▣  Stabilising Onchain Stablecoin Risks
- ▣  On Crypto-backed Loans

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_16

References (1/2)

- ▣ Adams, H., Zinsmeister, N., Robinson, D. (2020). [Uniswap v2 Core](#). Technical white paper.
- ▣ Andrews, D.W.K. (1993). [Tests for parameter instability and structural change with unknown change point](#). *Econometrica*, 61(4), 821–856.
- ▣ Athreya, K.B. (1987). [Bootstrap of the mean in the infinite variance case](#). *Annals of Statistics*, 15(2), 724–731.
- ▣ Bai, J. (1997). [Estimation of a change point in multiple regression models](#). *Review of Economics and Statistics*, 79(4), 551–563.
- ▣ Balke, N.S., Fomby, T.B. (1997). [Threshold cointegration](#). *International Economic Review*, 38(3), 627–645.
- ▣ Baur, D.G., Lucey, B.M. (2010). [Is gold a hedge or a safe haven? An analysis of stocks, bonds and gold](#). *Financial Review*, 45(2), 217–229.
- ▣ BlackRock (2026). [iShares Bitcoin Trust ETF \(IBIT\)](#). Product page.
- ▣ Bouri, E., Molnár, P., Azzi, G., Roubaud, D., Hagfors, L.I. (2017). [On the hedge and safe haven properties of Bitcoin: is it really more than a diversifier?](#). *Finance Research Letters*, 20, 192–198.
- ▣ Dickey, D.A., Fuller, W.A. (1979). [Distribution of the estimators for autoregressive time series with a unit root](#). *Journal of the American Statistical Association*, 74(366), 427–431.
- ▣ Dimson, E. (1979). [Risk measurement when shares are subject to infrequent trading](#). *Journal of Financial Economics*, 7(2), 197–226.
- ▣ Diop, P.O., Chevallier, J., Sanhaji, B. (2024). [Collapse of Silicon Valley Bank and USDC depegging: a machine learning experiment](#). *FinTech*, 3(4), 569–590.
- ▣ Efron, B. (1979). [Bootstrap methods: another look at the jackknife](#). *Annals of Statistics*, 7(1), 1–26.
- ▣ Fama, E.F., French, K.R. (2015). [A five-factor asset pricing model](#). *Journal of Financial Economics*, 116(1), 1–22.

References (2/2)

- ▣ Forbes, K.J., Rigobon, R. (2002). [No contagion, only interdependence: measuring stock market comovements](#). *Journal of Finance*, 57(5), 2223–2261.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Hansen, B.E. (1996). [Inference when a nuisance parameter is not identified under the null hypothesis](#). *Econometrica*, 64(2), 413–430.
- ▣ Koenker, R., Bassett, G. (1978). [Regression quantiles](#). *Econometrica*, 46(1), 33–50.
- ▣ Künsch, H.R. (1989). [The jackknife and the bootstrap for general stationary observations](#). *Annals of Statistics*, 17(3), 1217–1241.
- ▣ Kwiatkowski, D., Phillips, P.C.B., Schmidt, P., Shin, Y. (1992). [Testing the null hypothesis of stationarity against the alternative of a unit root](#). *Journal of Econometrics*, 54(1–3), 159–178.
- ▣ Liu, Y., Tsyvinski, A. (2021). [Risks and returns of cryptocurrency](#). *Review of Financial Studies*, 34(6), 2689–2727.
- ▣ Liu, Y., Tsyvinski, A., Wu, X. (2021). [Common risk factors in cryptocurrency](#). Accepted version, 6 June 2021, SSRN 3379131.
- ▣ Liu, Y., Tsyvinski, A., Wu, X. (2022). [Common risk factors in cryptocurrency](#). *Journal of Finance*, 77(2), 1133–1177.
- ▣ Lyons, R.K., Viswanath-Natraj, G. (2023). [What keeps stablecoins stable?](#). *Journal of International Money and Finance*, 131, 102777.
- ▣ Milionis, J., Moallemi, C.C., Roughgarden, T., Zhang, A.L. (2022). [Automated market making and loss-versus-rebalancing](#). arXiv:2208.06046.
- ▣ Newey, W.K., West, K.D. (1987). [A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix](#). *Econometrica*, 55(3), 703–708.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#). *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Trimborn, S., Härdle, W.K. (2018). [CRIX an index for cryptocurrencies](#). *Journal of Empirical Finance*, 49, 107–122.